

Dialogue Systems Lab 4 Part A and Part A-VG

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For Part A, I tested scientific names for plants, an uncommon name, fictional names, real people names, and a non-English real location name. When it comes to the fictional name Dumbledore, the recognizer got it right at first but with low confidence (0.37035665), but later attempts produced “dumbbell” and “door” as well. However, for those later attempts I tried to adjust my pronunciation to what I thought would be more accurate, but the outputs showed that my pronunciation at first attempt made Dumbledore more likely to get recognized by the automatic speech recognizer. This suggests that accent and pronunciation may affect the quality of the recognition. On the other hand, the fictional name Voldemort was recognized correctly several times with high confidence scores (approximately ranging between 0.82 and 0.90), so this suggests that some fictional characters are either easier for the system to recognize or harder for me to pronounce. When it comes to the non-English real location name, I tested the Cypriot village name Kakopetria, and in every attempt the system misrecognized it as “Cacopatria” or “Cacophetria” with very low confidence (approximately ranging between 0.05 and 0.07), which could suggest that uncommon place names influenced by another language are difficult for the recognition system.

When it comes to scientific names of plants, I tested Chrysanthemum. The recognizer usually recognized it with moderate confidence (approximately ranging between 0.54 and 0.69), except one instance when it misrecognized it as “Santhemom” with very low confidence (approximately 0.05). I also briefly tested Magnolia liliflora similar to the example in the instructions, but the recognizer tended to separate it into more familiar English words (“Magnolia Lily flora”) with low confidence (approximately 0.30). I also tested the classical composer’s name Tchaikovsky, and it was recognized in the first attempt, however the confidence was low to moderate (approximately 0.38), which probably had to do with the non-English name and my pronunciation. When it comes to real people’s names, Vlad was recognized with moderate confidence (approximately 0.46), and the name Lena was recognized several times, however the confidence levels remained very low throughout the attempts (approximately ranging between 0.09 to 0.16). This suggests that even simple personal names may be recognized correctly while the confidence levels of the system remain low.

I inspected the ASR output in the browser console. The scores varied: Voldemort was consistently correctly recognized with high confidence, while the consistent correct recognitions of Lena, Vlad and Tchaikovsky had low or moderate confidence, and Chrysanthemum and Dumbledore were more inconsistent but also did not have high confidence when they were successfully recognized. Kakopetria and Magnolia liliflora were never recognized successfully and had low confidence. I think that recognition failed because of accent, pronunciation differences, uncommon vocabulary, and language mismatch. To improve this, the model could be trained with difficult names and words, the system could ask the user to repeat low-confidence words, or it could ask the user to confirm recognitions with low confidence.

After training the custom speech model for Part A-VG, the recognition performance improved for both the common and the uncommon words, with all the words being recognized correctly and most words having high confidence scores. The only exception to this is Chrysanthemum, which was recognized correctly at every attempt, but had a moderate confidence score at the first attempt of approximately 0.55. However, I attempted again with clearer pronunciation and the confidence increased to approximately 0.96. The names Vlad and Lena had confidence scores approximately ranging from 0.84 to 0.94 and 0.84 to 0.91 respectively, which were significant increases, especially for Lena. The fictional names Voldemort and Dumbledore had

confidence scores approximately ranging between 0.89 to 0.91 for Voldemort and 0.97 for Dumbledore, showing a slight increase for Voldemort but a significant increase for Dumbledore. The uncommon name Tchaikovsky had a significant increase as well, with the confidence score being approximately 0.98. Finally, the names that could not be correctly recognized during part A, Magnolia liliflora and Kakopetria, were correctly recognized with high confidence scores (approximately 0.88 for Kakopetria and approximately 0.98 for Magnolia liliflora). The results suggest that the custom speech model improved recognition accuracy and recognition confidence levels, while pronunciation quality still affected confidence levels.